ABT RECERTIFICATION EXAMINATION #20

May 5, 2003

PART A

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1994 and 1999. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 572-5525 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following is INCORRECT?
   A) when ingested, the seeds of wisteria (*Wisteria floribunda*) and castor bean (*Ricinus communis*) cause gastrointestinal disturbances
   B) *Philodendron* and *Dieffenbachia* both belong to the Arum family and cause allergic dermatitis
   C) while taxanes from the European yew can cause cardiovascular toxicity, paclitaxel from the Pacific yew has chemotherapeutic activity
   D) excitatory amino acids are found in algae, fungi, and higher plants, and they mimic natural neural transmitters
   E) pyrrolizidine alkaloids are the causative agents of liver damage from the ingestion of rattle box (*Crotalaria*), heliotrope (*Helotropium*), and comfrey (*Symphytum*)

2. The toxins of the tansy ragwort (*Senecio jacobea*) induce necrosis of the
   A) stomach
   B) small intestine
   C) liver
   D) kidney
   E) C and D

3. In general, the composite safety factor used by the Food and Drug Administration when using the results of animal experimental data to calculate Acceptable Daily Intakes (ADIs) for food additives is
   A) 10
   B) 100
   C) 1,000
   D) 10,000
   E) 100,000

4. The Food and Drug Administration generally considers that carcinogens migrating from food packaging materials pose a negligible risk at what level?
   A) 0
   B) 1 in 100,000
   C) 1 in 1,000,000
   D) 1 in 10,000,000
   E) none of the above

5. Which of the following is true about organotin toxicity?
   A) both trimethyltin and triethyltin are neuronopathy-inducing agents
   B) both trimethyltin and triethyltin are myelinopathy-inducing agents
   C) both trimethyltin and triethyltin are axonopathy-inducing agents
   D) trimethyltin is a neuronopathy-inducing agent, and triethyltin is a myelinopathy-inducing agent
   E) tributyltin is exclusively a neurotoxic agent

6. For which of the following are the major toxic effects described as acute chemical pneumonitis, hypersensitivity, chronic granulomatous pulmonary disease and malignant lung tumors?
   A) lead
   B) mercury
   C) beryllium
   D) cobalt
   E) molybdenum

7. Which of the following statements is true regarding oxalate poisoning?
   A) calcium oxalate of plant origin is absorbed unchanged
   B) sodium oxalate of plant origin is absorbed and causes hypercalcemia
   C) potassium oxalate of plant origin is absorbed and converted to an insoluble form
   D) oxalates of plant origin primarily cause liver damage
   E) plants do not contain either potassium or sodium oxalates

8. Solvent-induced chronic encephalopathy
   A) is characterized by well defined cortical lesions
   B) has been established to occur in painters
   C) is a controversial disease characterized by neurological symptoms and functional deficits
   D) manifests as cognitive deficits and neurodegenerative changes
   E) has been established to occur only from exposure to multiple solvents

9. Which of the following is correct?
   A) carbon monoxide is soluble in water
   B) CO shifts oxygen/hemoglobin dissociation curve to the right
   C) CO affinity for hemoglobin = 210 times that of O_2
   D) smokers have approximately the same levels of COHb as nonsmokers
   E) none of the above

10. Which of the following is/are true of nitrogen dioxide?
A) it is a deep lung irritant
B) it causes pulmonary edema in humans
C) it is the agent involved in silo fillers' disease
D) A and B
E) all of the above

11. The deposition of asbestos fibers in the respiratory tract occurs mainly by
A) interception
B) impaction
C) sedimentation
D) diffusion
E) no particular deposition process predominates

12. Exposure to ozone in low concentrations produces alterations in respiration characterized by all of the following EXCEPT
A) shallow breathing
B) decreased pulmonary compliance
C) rapid breathing
D) increase in CO diffusion capacity
E) airway hyperactivity

13. The rodenticide sodium fluoroacetate exerts its toxic effect by which one of the following mechanisms?
A) irreversibly binds to and inhibits acetylcholinesterase
B) forms a peroxy radical that rapidly depletes cellular glutathione
C) is metabolized to fluorocitrate that inhibits aconitase, blocking the citric acid cycle
D) is rapidly dehalogenated, resulting in an acute fluoride-induced renal failure
E) binds to cytochrome oxidase and inhibits mitochondrial electron transport

14. Saccharin has been found to enhance tumor development in
A) rat urinary bladder
B) rat lung
C) mouse liver
D) all of the above
E) none of the above

15. Which of the following is/are true regarding toxic gases?
A) the dominant driving force in the uptake of toxic gases is diffusion
B) the primary force for diffusion is the concentration in the inspired air
C) the solubility of the gas in water is the major characteristic determining the regional toxicity of gases
D) all of the above
E) none of the above

16. Which of the following statements accurately characterizes the developmental or reproductive toxicity of glycol ethers?
A) longer chain glycol ethers such as 2-butoxyethanol are stronger teratogens than short chain compounds such as 2-methoxyethanol
B) epidemiology studies in workers have failed to demonstrate the adverse effect of 2-methoxyethanol on the male reproductive system
C) 2-methoxyethanol produces teratogenic effects in rats including malformations of the heart and skeletal system
D) the acetates of the corresponding ethers do not produce similar teratogenic effects
E) the developmental toxicity produced by glycol ethers is route specific; i.e., produced through inhalation but not dermal exposure

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

17. Centrilobular hepatocyte necrosis is a primary lesion of
1) carbon tetrachloride
2) allyl alcohol
3) ethanol
4) hexylene glycol

18. Which of the following solvents covalently modify(ies) proteins either directly or through a metabolite?
1) bromobenzene
2) hexane
3) benzene
4) carbon disulfide

19. What types of toxicity has/have been reported for carbon disulfide?
1) neurotoxicity
2) cardiovascular toxicity
3) hepatotoxicity
4) renal toxicity

20. Renal toxicity in experimental animals is associated with exposure to
1) bromobenzene
2) carbon tetrachloride
3) 1,2-dichloroethylene
4) chloroform

21. Lindane
1) has been used to treat head lice in children
2) has been used in dog flea and tick dips
3) is the gamma isomer of hexachlorocyclohexane
4) has high mammalian toxicity because of its slow metabolism

22. Repeated exposure to anticholinesterase organophosphorus insecticides may produce the development of tolerance to their toxicity as a result of
1) rapid acetylcholinesterase resynthesis
2) reduced nicotinic acetylcholine receptor sites
3) nonspecific binding of insecticides to the serine-containing enzymes other than acetylcholinesterase
4) reduced muscarinic acetylcholine receptor sites

23. Chronic exposure to low doses of dichloropdihenltrichloroethane (DDT)
1) can be expected to decrease the acute toxicity of DDT because of induction of enzymes utilized in DDT metabolism
2) decreases the acute toxicity of administered parathion because of inhibition of enzymes necessary for the metabolism to paraoxon
3) has been shown experimentally to increase the acute toxicity of administered parathion because of induction of oxidative enzymes
4) enhances the chronic toxicity of phenobarbital

24. Occupational exposure to kepone has resulted in symptoms of poisoning that persist after removal of workers from exposure. Possible reason(s) include
1) cleavage of the glucuronide in the gut by intestinal flora allows reabsorption of the toxicant into the bloodstream
2) repair of toxicant-induced damage to the nervous system is very slow
3) hepatotoxicity induced by the toxicant has reduced the ability of the liver to perform phase II conjugation
4) the toxicant is slowly released from depots in the bone

25. Toxicologic effects produced by pyriminil, a rodenticide, include
1) destruction of the B islet cells in the pancreas
2) peroxisomal proliferation
3) neurologic impairment
4) inhibition of spermatogenesis

26. Which of the following statements is/are correct?
1) cacodylic acid, an organic arsenical, and mono- and disodium methyl arsenate are used as pesticides
2) the toxicity of cacodylic acid resembles that of trivalent or inorganic arsenicals
3) the treatment of poisoning by cacodylic acid is similar to inorganic arsenicals
4) the toxicity of cacodylic acid is generally greater than that of the inorganic arsenicals

27. Which of the following is/are correct?
1) a major effect of benzene is hematopoietic toxicity and this effect is unique to benzene among the monocyclic aromatic hydrocarbons
2) one component of acute human poisoning due to ingestion of ethylene glycol results from its metabolism to glycolic acid and consequent development of profound acidosis
3) the International Agency for Research on Cancer has concluded that, whereas the evidence for the carcinogenicity of ethanol and alcoholic beverages in animals was inadequate, there is "sufficient evidence" for the carcinogenicity of alcoholic beverages in humans that include tumors of the oral cavity, pharynx, larynx, esophagus and liver
4) the target for acute methanol toxicity is the liver

28. Exposure to the rodenticide zinc phosphide results in
1) central nervous system stimulation and secondary paralysis
2) hemolysis
3) dyspnea, general weakness and tremors and perhaps coma with convulsions
4) aplastic anemia

29. Which of the following is/are correct with regard to insecticides?
1) many of the chemical insecticides in use today are neurotoxicants and act by poisoning the nervous systems of the target organism
2) the highly lipid soluble nature of the organochlorine insecticides results in their accumulation in body tissues having a high lipid content
3) the mechanism by which organophosphorus and carbamate ester insecticides elicit their toxicities is identical and is associated with the inhibition of nervous tissue acetylcholinesterase
4) polychlorinated biphenyls are organochlorine insecticides

30. Which of the following types of reactions can radionuclides undergo?
1) alpha emission
2) beta emission
3) isomeric transition
4) electron capture

31. In lead-induced anemia
1) the red blood cells are microcytic and hypochromic, and usually there are increased numbers of reticulocytes with basophilic stippling, which results from inhibition of the enzyme pyrimidine-5-nucleotidase (Py-5-N)
2) inhibition of Py-5-N activity and nucleotide accumulation with lead exposure affects erythrocyte membrane stability and survival by altering cell energy metabolism
3) an effect on heme synthesis is inhibition of ALA-D (aminolevulinic acid dehydratase), resulting in a negative exponential relationship between ALA-D and blood lead
4) there is an inverse relationship between Py-5-N activity with blood lead concentration

32. Workers exposed to cobalt-containing dust may develop
1) pneumoconiosis
2) allergic dermatitis
3) bronchial asthma
4) peripheral neuropathy

33. Asbestos fibers differ from man-made mineral fibers (MMMF) in which of the following ways?
1) they are crystalline in nature while MMMF are vitreous
2) they cause upper respiratory disease and MMMF do not
3) they tend to break parallel to the length of the fiber while MMMF tend to break crosswise
4) they can be either serpentine or amphibole while MMMF are usually serpentine

34. Radon
1) is of concern as an indoor air pollutant
2) has been associated with leukemia in uranium mine workers
3) is the greatest contributor to natural (background) ionizing radiation
4) is a long-lived decay product of radium

35. Human exposure to arsenic is known to cause
1) skin cancer
2) skin lesions including hyperkeratosis and hyperpigmentation
3) lung cancer
4) peripheral vascular disease which can sometimes be so severe as to progress to gangrene

36. Which of the following drug(s) has/have been shown to be cardiotoxic?
1) anthracyclines
2) tricyclic antidepressants
3) digitalis
4) 5-fluorouracil

37. Venoms of the scorpion fishes
1) are extremely heat stable
2) can cause hypotension, headaches, seizures and paralysis
3) interfere with blood clotting
4) cause intense pain and are heat labile

38. All of the following plants contain digitalis-like glycosides EXCEPT
1) foxglove (*Digitalis purpurea*)
2) lily-of-the-valley (*convallaria majalis*)
3) yellow oleander (*Thevetia peruviana*)
4) jimson weed (*Datura stramonium*)

39. Which of the following statements about lead intoxication is/are correct?
1) children generally absorb more ingested lead than adults
2) shortly after exposure, almost all the lead in blood is found in red blood cells
3) the nervous system is a major target organ for lead
4) blood lead levels in excess of 10 micrograms/deciliter in children are believed to be associated with the impairment of cognitive and behavioral development

40. Which of the following is/are characteristic of radiation?
1) it may be electromagnetic
2) it may be particulate
3) neutrons have high kinetic energy and penetrate readily
4) single source alpha particles show a discrete, not a continuous, energy distribution

41. Which of the following is INCORRECT regarding acute, elemental iron posioning?
1) the first signs of poisoning include hematemesis, abdominal pain, and bloody diarrhea
2) necrosis of gastric mucosa results from the direct corrosive effects of iron
3) metabolic acidosis and shock may occur
4) renal necrosis may ultimately develop

42. When conducting a nonclinical toxicity study in support of a pharmaceutical development program, which of the following distinguish(es) a study in which the test article is a recombinant protein from a study in which the test article is categorized as a "small molecule"?
1) the conduct of a study with a recombinant protein does not need to adhere to Good Laboratory Practices
2) analysis of the test article stability and verification of accuracy of formulation are not critical components of a study with a recombinant protein
3) it is necessary to test recombinant proteins in a nonhuman primate species
4) it is usually important to characterize the potential for the recombinant protein to elicit an antibody response in the test species

43. The Food Quality Protection Act of 1996 requires the Environmental Protection Agency to
1) assess cumulative risks to human health that may result from exposure to pesticides and other substances that are toxic by a common mechanism of action
2) determine how to assess pesticides with common mechanisms of action
3) describe data requirements and specifications for their use in reaching conclusions
4) evaluate the specific susceptibility of infants and children to toxic effects of pesticides

44. Which of the following statements regarding manganese is/are correct?
1) toxicity occurs primarily as a result of inhalation from occupational exposure
2) it is an essential dietary nutrient for man
3) it can produce a Parkinson's-like syndrome with chronic exposure
4) CNS toxicity is readily reversed by chelation therapy

45. The safety of a food contact (indirect) additive such as a constituent food packaging material is
1) not of concern because this constituent is not purposefully added to food
2) in part, determined by calculation of exposure following extraction in one or more types of food-simulating media
3) determined by macerating the packaging material and feeding it to two species of rodents
4) of concern if the constituent is shown to be carcinogenic and present at levels exceeding an estimated risk of 1 in 1,000,000

46. Possible signs of toxicity in response to chronic exposure to a potent corticosteroid would likely include which of the following?
1) suppression of the immune system
2) hyperglycemia resulting from changes in carbohydrate metabolism
3) developmental abnormalities in the fetuses of the exposed animal
4) leukemia

47. Which of the following statements is/are true regarding the role of muscarinic receptors and pharmacologic effects of antimuscarinic agents?
1) tissues in which muscarinic receptors are found include the cardiovascular system, gastrointestinal system and urinary bladder
2) basic functions of muscarinic receptors are mediated by interactions with members of the family of G proteins
3) scopolamine has marked CNS effects at low therapeutic doses
4) atropine increases salivation and sweating

48. Phenobarbital has been shown to induce thyroid tumors in chronic rodent bioassays. Which of the following is/are correct?
1) it has been shown to block the uptake of iodine from the blood and into the thyroid follicular lumen
2) the tumors produced are regarded as not predictive in estimating the risk to humans
3) it acts by inhibiting the synthesis of T3, T4
4) it results in an increase in the biliary excretion of T4 glucuronide conjugates

49. Zearolenones
1) are produced by *Fusarium spp.*
2) decrease the reproductive potential of livestock, especially in swine
3) cause a hyperestrogenic syndrome, especially in swine
4) cause ergotism

50. Which of the following is/are true of cadmium?
1) toxicity is produced by binding to target tissue sulfhydryl groups
2) liver toxicity may be prevented by binding to sulfhydryl groups of metallothionein
3) kidney toxicity may be enhanced by formation of a complex with metallothionein in the liver which is then transported to the kidney
4) storage in the liver may reduce toxicity to the testes

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Key for the following questions
Choose the single most appropriate answer

1. The tissue site of toxicant induced necrosis may be dependent on
   A) the excretory pathway of the agent
   B) the tissue's ability to metabolize the agent
   C) different tissue concentrations of the agent
   D) A, B, and C
   E) A and B

2. The drug of choice in the treatment of acute nickel carbonyl poisoning is
   A) dithiocarb (diethyldithiocarbamate)
   B) desferroxamine
   C) BAL (British anti-lewisite)
   D) n-acetylcystein
   E) 2-PAM

3. Which of the following is true about inhalation of phosgene?
   A) chronic inhalation results in mandibular necrosis
   B) inhalation causes necrosis and inflammation of the small airways and alveoli that may lead to noncardiogenic pulmonary edema
   C) high level exposure causes bronchodilation
   D) acute inhalation results in nausea, vomiting, dizziness and confusion
   E) acute inhalation causes nausea, euphoria, ataxia, dizziness, agitation and coma

4. A retrospective study is characterized by all of the following EXCEPT
   A) it is less expensive than a prospective study
   B) relative risk may be estimated
   C) incidence rates may be computed
   D) the controls selected must be disease-free
   E) past exposures levels are easy to quantify

5. What is the most likely reason for hospital admission following the ingestion of C_9-C_{15} hydrocarbons?
   A) CNS depression
   B) chemical pneumonitis
   C) nephrotoxicity
   D) myocardial sensitivity to catecholamines
   E) severe hepatic damage

6. Incidence is defined as
   A) total number of cases
   B) number of new cases
   C) number of existing cases
   D) number of cases/population at risk
   E) number of new cases/year

7. Diethylstilbestrol is eliminated mainly in which of the following?
   A) bile
   B) urine
   C) expired air
   D) saliva
   E) milk

8. Which of the following groups contain examples, of which all are Phase I biotransformation enzymes?
   A) epoxide hydrolase, UDP-glucuronosyltransferase, cytochrome P450, FAD monooxygenase
   B) epoxide hydrolase, UDP-glucuronosyltransferase, cytochrome P450, glutathione S-transferase
   C) epoxide hydrolase, cytochrome P450, FAD monooxygenase, alcohol dehydrogenase
   D) UDP-glucuronosyltransferase, glutathione S-transferase, sulfotransferase
   E) none of the above

9. What is the correct dose volume for a 250 g rat if the target dose level is 5000 mg/kg and the test substance is dissolved in the vehicle at a concentration of 25% w/v?
   A) 2 ml
   B) 5 ml
   C) 10 ml
   D) 20 ml
   E) 7.5 ml

10. Threshold Limit Values
   A) serve as an index of relative toxicity for industrial chemicals
   B) if followed, protect all workers following repeated exposure for a working lifetime
   C) are recommended for use in the evaluation or control of community air pollution
   D) are established on chemical specific information that may differ from substance to substance
   E) when rigorously applied, define the line between safe and dangerous concentrations

11. There is a gradient of cytochrome P450 enzyme activity across the liver acini. Which of the following is correct?
   A) greater amounts of constitutive cytochrome P450 enzymes occur in the centrilobular region
   B) greater amounts of constitutive cytochrome P450 enzymes occur in the periportal region
   C) the activity gradient across the acinus is positively correlated to an oxygen gradient
   D) the activity gradient is related to differences in regulatory or binding protein expression
   E) the activity gradient is related to the tissue concentration of xenobiotic at the specific region

12. Six ml of a chemical the elimination of which follows zero-order kinetics was ingested. Plasma concentration of the chemical became one-half of the initial concentration 3 hours after ingestion. The additional time needed for elimination of 95% of the chemical ingested is
A) 1 hour
B) 3 hours
C) 9 hours
D) 18 hours
E) 36 hours

13. Which of the following approaches is currently preferred when attempting to determine if exposure to a chemical may result in adverse health impacts?
A) chemical residue determination in samples of environmental media
B) chemical residue analysis of tissues from biological receptor
C) identification and quantification of biomarkers of effect
D) obtaining additional information from the source of the released chemical
E) determination of the half-life of the chemical in the environment

14. What is the median value of the following series of numbers 23,7,15,1,7,5,27,20,11,4?
A) 10
B) 9
C) 7
D) 26
E) 12

15. Shortcomings in the No Observed Adverse Effect Level (NOAEL)/safety factor approach for food safety assessment include all of the following EXCEPT
A) once the NOAEL is identified, the rest of the dose-response curve is ignored
B) the NOAEL, by definition, has to be one of the experimental test dosages
C) the NOAEL will vary depending on experimental design and dose selection which results in an Acceptable Daily Intake (ADI) that does not reflect the same level of risk
D) experiments that use fewer test animals result in lower NOAELs
E) the variability in the derived ADI is highly dependent on the inherent variability of the NOAEL approach

16. Which of the following types of preclinical studies is generally NOT conducted prior to proceeding with single dose human Phase I clinical trials in the U.S.?
A) genetic toxicology screen
B) fertility test
C) pharmacokinetic information
D) a general toxicology study in a rodent species
E) a general toxicology study in a nonrodent species

17. Assume the following (1) a long-term animal study identifies a no-observed-adverse-effect-level of 10 mg/kg/day, (2) a human weighs 70 kg, and (3) an uncertainty factor of 100. What is the Acceptable Daily Intake in humans calculated from the animal study?
A) 10 mg/day
B) 7 mg/day
C) 3 mg/day
D) 10 mg/kg
E) 7 mg/kg

18. The persistence of organophosphorous and carbamate insecticides in soils is influenced by
A) binding to the soil
B) water content of the soil
C) microflora present
D) A, B and C
E) A and C

19. According to the U.S. Environmental Protection Agency, male rat renal tubule tumors arising as a result of a process involving alpha$_2$u-globulin
A) have been associated with urinary bladder neoplasia
B) are included in dose-response extrapolations for human carcinogenic risk assessment
C) tend to regress over long recovery periods
D) are not considered to indicate a potential human carcinogenic hazard
E) are related to genotoxicity

20. Which is true regarding the toxicological profile of hazardous wastes?
- A) generally, waste substances are simple chemical mixtures of low order toxicity and the toxicology has been well-defined
- B) the organ system toxicology and human exposure outcomes can be predicted by reviewing the literature for the most toxic substances in the mixture
- C) the toxicological profile can be determined by conducting dosing studies in multiple animal species because the hazardous waste generally is consistent in chemical composition
- D) hazardous waste is generally a highly complex mixture of chemicals, the toxicity of which may be difficult to predict based on partial chemical analyses
- E) hazardous waste is generally a highly complex mixture of chemicals, the toxicity of which may be accurately predicted based on partial chemical analyses

21. Toxicants may pass through a membrane via numerous mechanisms. Which of the following statements is true for an active transport system?
- A) requires an expenditure of energy
- B) cannot transport polar compounds
- C) system is not selective
- D) does not move against electrochemical gradients
- E) transport maximum is not exhibited

22. Which of the following is NOT a plausible explanation for the substantial decrease in serum albumin concentration observed during a toxicity study?
- A) the agent being tested has caused extensive damage to the hepatic parenchyma
- B) the agent being tested has caused the albumin protein to aggregate, resulting in its accumulation within the splenic vasculature
- C) the albumin has extravasated due to vascular leakage
- D) damage to the glomerular apparatus in the kidney has permitted extensive elimination in the urine
- E) an "acute phase" response in the liver has resulted in down-regulation of albumin synthesis

23. Which of the following best describes the tolerance distribution model for low dose extrapolation for carcinogenesis?
- A) the tumor response is characterized by a single hit within a specified time interval and the probability of that hit follows a Poisson distribution
- B) the tumor response requires that multiple steps of differentiation be affected and that a fixed probability is associated with each step
- C) a linear dose response for tumors is based upon the delivered dose to a target organ, which includes all animals at risk
- D) the applied dose, time of the appearance of a tumor and the time to death predict the probability of outcome
- E) each member of a population has a tumor threshold level below which that individual will not respond

24. Probability distributions in risk assessment are most useful in
- A) deterministic risk assessment approaches
- B) qualitative uncertainty analysis
- C) log-probit modeling of carcinogenic responses
- D) Monte Carlo simulations of exposure estimation
- E) residual risk calculations

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

25. Breathing 100% oxygen is effective in treating intoxication with
- 1) sodium nitrite
- 2) 2,4-dinitrophenol
- 3) acetaminophen
- 4) carbon monoxide

26. Which of the following is/are correct?
- 1) cholinesterases (ChEs) are broadly classified as either acetylcholinesterase (true cholinesterase, or AChE), or butyrylcholinesterase (BuChE, plasma cholinesterase, or pseudocholinesterase)
- 2) plasma ChE levels may be depressed in parenchymal liver disease and malnutrition
- 3) plasma ChE is synthesized in the liver but its physiologic function is obscure
- 4) organophosphate insecticides are potent inhibitors of both AChE and BChE

27. Which of the following enzymes is/are involved in the synthesis of mercapturic acid derivatives from xenobiotic-glutathione conjugates?
1) cysteine conjugate beta-lyase
2) cysteine conjugate N-acetyl transferase
3) rhodanese
4) gamma glutamyl transpeptidase

28. Which of the following apply to delayed neurotoxicity from tri-o-cresyl phosphate (single dose)?
1) the chicken is sensitive but the rat is not
2) very young animals are generally insensitive
3) the cat is sensitive but the rabbit is not
4) the amount administered is important

29. Which of the following statement(s) is/are true regarding the susceptibility of infants to nitrate induced methemoglobinemia?
1) infants have a less acidic gastric environment than adults
2) infants have a lower enzymatic capacity of methemoglobin reductase than adults
3) fetal hemoglobin is more susceptible to oxidation by nitrite than adult hemoglobin
4) infant water intake per body weight is greater than that of adults

30. The mechanism of action for the hepatocarcinogenic effect of di(2-ethylhexyl)phthalate (DEHP) includes
1) peroxisome proliferation by its metabolite, mono(2-ethylhexyl phthalate)
2) peroxisome proliferation directly by DEHP
3) a threshold-based response
4) genotoxicity

31. Which of the following is/are correct?
1) the Animal Welfare Act (AWA) gives the force of law to minimum standards for the care of research animals
2) unlike the AWA, the Public Health Service (PHS) Policy on Humane Care and Use of Laboratory Animals covers all vertebrates
3) the Association for Assessment and Accreditation of Laboratory Animal Care, International (AAALAC) provides a voluntary accreditation process for research animal facilities
4) the NIH "Guide for the Care and Use of Laboratory Animals" provides recommended standards for research institutions to ensure appropriate and humane care of laboratory animals

32. Metabolism of foreign compounds may be influenced by which of the following?
1) species variations
2) age of experimental animals
3) the type of bedding
4) the lighting schedule used the animal room

33. Which of the following statements is/are true concerning information used in the regulatory process?
1) epidemiology, controlled clinical exposures, animal toxicology, *in vitro* studies and structure activity relationships are employed
2) lack of exposure information often limits the conclusions that can be made from epidemiology studies
3) uncertainties are inherent in extrapolating from animals to humans and from high-dose animal exposures to low-dose human exposures
4) animal toxicology studies outweigh all other sources of information

34. Which of the following is/are correct?
1) dogs do not usually N- and O- acetylate xenobiotics by enzyme systems similar to rodents and humans
2) cats have a low capacity to glucuronidate xenobiotics
3) cats are more susceptible to acetaminophen toxicity than humans
4) cats and dogs exhibit significant differences in susceptibility to the toxic effects of phenols because of differences in the ability to glucuronidate

35. Ideally bioassays for the carcinogenic effects of chemicals should achieve which of the following?
1) at least 25% of the animals should survive to termination
2) administration should be by a route relevant to human exposure
3) the strains used should have a well established background of spontaneous neoplasms
4) the group size should be at least 50 animals per sex to permit detection of neoplasms at an incidence of 5-10%

36. Which of the following types of toxicological information is/are considered "continuous data"?
1) histopathological findings
2) body weights
3) mortality data
4) clinical chemistry/hematology data

37. Which of the following parent compounds is/are correctly paired with its ultimate toxicant?
1) amygdalin - HCN
2) acetaminophen - N-acetyl-*p*-benzoquinoneimine
3) ethylene glycol - oxalic acid
4) benzo[*a*]pyrene (BP) - BP-7,8-diol-9,10-epoxide

38. Which of the following is/are true concerning sulfur dioxide?
1) asthmatic individuals have increased sensitivity
2) increased airway resistance can occur after brief exposure
3) exercise during exposure enhances its effect in normal individuals
4) it is a direct acting pulmonary carcinogen

39. Static acute aquatic toxicity tests are useful for
1) deriving estimates of upper limits of concentrations producing toxic effects
2) developing understanding of the concentration-response relationship
3) evaluating relative sensitivity of different organisms to the same material
4) determining potential effects of chronic exposure

40. Identify the correct clinical chemistry parameter with the associated organ.
1) creatine kinase - heart
2) creatinine and urea nitrogen - kidney
3) alanine aminotransferase - liver
4) amylase - pancreas

41. Occupational asthma
1) occurs in garment workers exposed to formaldehyde
2) is caused by exposure to toluene diisocyanate
3) is associated with acetylene exposure
4) occurs in workers exposed to detergents containing enzymes

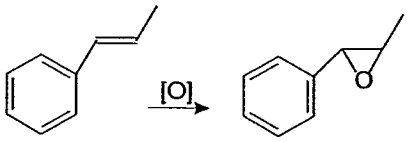
42. Which of the following is/are correct?
1) clearance (CL) is equal to the area under the plasma concentration curve (AUC) divided by dose (D), i.e. CL=AUC/D
2) total serum concentration for substances that are highly protein bound may not be reflective of toxicological or biological activity
3) AUC, but not maximum concentration (Cmax), is important when comparing toxicity by intravenous and oral administration
4) pharmacokinetic data from animals given toxic doses may be of little value when extrapolating to the human situation at lower doses

43. Urine analysis can be used for biological monitoring of exposure to
1) lead
2) cadmium
3) vinyl chloride
4) carbon monoxide

44. Which of the following is/are true?
1) most chemicals are eliminated by first-order processes
2) biotransformation and active transport processes, as well as protein binding, have unlimited capacities in most common laboratory species
3) prolonged residency time of a compound in the body occurs during the transition from first-order to saturation kinetics
4) the elimination of ethanol after intoxication follows first-order kinetics

45. Which of the following chemical pairs is/are appropriate?
1) propanol-carbon tetrachloride: potentiation
2) morphine-naloxone: synergism
3) mercury-BAL: chemical antagonism
4) propoxyphene-amytal: dispositional antagonism

Figure 30006



46. The reaction shown above in figure 30006 illustrates the formation of a reactive metabolite by
1) phase I metabolism
2) a reaction catalyzed by a cytosolic enzyme
3) oxidative biotransformation
4) hydrolytic biotransformation

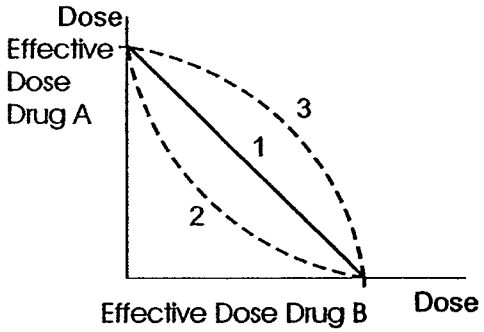
47. Which of the following statements about physiologically based pharmacokinetic (PBPK) models is/are true?
1) PBPK uses mass balance equations to model organ and tissue physiology and biochemical characteristics
2) if first order kinetics apply, a plot of natural log of plasma concentration versus hours after exposure will have a constant slope
3) partition coefficients for tissue compartments must be known or estimated for the chemical
4) transport of the chemical is assumed to be instantaneous between the blood and perfused tissues when using diffusion limited models

48. A drug is a weak acid with a dissociation constant of 10^{-5}; the drug is administered to a patient in a strongly buffered solution with a pH of 8. Which of the following is/are correct?
1) the drug has a pKa of 5
2) drug absorption in the stomach will be decreased when administered in the buffered solution
3) more of the drug will be ionized at pH 8 than at stomach pH
4) at pH 5, most of the drug will be ionized

49. Which of the following statements accurately characterize food consumption patterns in rodents as they grow older?
1) younger animals consume more food per unit of body weight than older animals
2) older animals consume more food per unit of body weight than younger animals
3) if diets containing the same concentration of test material are fed throughout a chronic study, animals will receive less test material per unit of body weight at the end of the study than at the beginning
4) if diets containing the same concentration of test material are fed throughout a chronic study, animals will receive more test material per unit of body weight at the end of the study than at the beginning

Key for the following questions
Choose the single most appropriate answer

Figure 30004



50. Refer to Figure 30004
Line 3 in the isobologram shown above depicts what type of toxicological interactions between Drug A and Drug B ?
A) antagonism
B) potentiation
C) synergism
D) additivity
E) margination

ABT RECERTIFICATION EXAMINATION #20

May 5, 2003 ***PART B***

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1994 and 1999. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 572-5525 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following is located in the renal cortex?
   A) collecting ducts
   B) descending loop of Henle
   C) glomerulus
   D) ascending loop of Henle
   E) distal tubule

2. Which of the following is NOT characteristic of anaplastic cells?
   A) cellular pleomorphism
   B) hypochromatic nuclei
   C) poorly differentiated
   D) increased nucleolar to nuclear ratio
   E) atypical mitotic figures

3. The tumor suppressor gene p53 is characterized by all of the following EXCEPT
   A) encodes a transcription factor protein
   B) contributes to monitoring and repair of DNA damage
   C) arrests cell growth in M-phase
   D) is inactivated by mutation in many human tumors
   E) promotes apoptosis

4. Which of the following are effects of inhaled mineral fibers in animals?
   A) parenchymal lung inflammation, pleural adenomas, nasal tumors
   B) parenchymal lung mesothelioma, pleural inflammation, pleural plaques
   C) pleural mesothelioma, parenchymal lung inflammation, pleural inflammation
   D) peritoneal tumors, colon tumors, lymphoma
   E) pleural inflammation, pleural plaques, pleural adenomas

5. Which of the following is NOT true regarding base excision repair?
   A) involves the actions of an N-glycosylase, apurinic/apyrimidinic endonuclease, exonuclease, polymerase and ligase
   B) is also known as "small patch" repair
   C) is characterized by enzymes specific for different kinds of altered bases
   D) involves the removal of more than 20 nucleotides
   E) none of the above

6. Which of the following is correct?
   A) rodents are used for developmental neurotoxicity studies because the nervous system in humans and rodents is mature at birth and has similar periods of susceptibility
   B) exposure of the human embryo and fetus to ethanol results in both structural and functional abnormalities of the nervous system
   C) the toxicity of methyl mercury in the fetal nervous system has only been demonstrated in nonhuman primates and humans
   D) injury to the developing nervous system is permanent because there are no compensatory mechanisms in the brain
   E) methyl bromide is a known neurotoxicant to the developing nervous system in experimental animals

7. Which of the following produces renal tubular necrosis and myocardial sensitization to sympathomimetics?
   A) carbon monoxide
   B) natural gas
   C) propylene glycol
   D) carbon tetrachloride
   E) carbon disulfide

8. Idiosyncratic injury to the liver is associated with
   A) chlorpromazine
   B) phenytoin
   C) erythromycin
   D) all of the above
   E) none of the above

9. Which segment of the nephron is the site of greatest organic anion transport activity?
   A) collecting duct
   B) glomerulus
   C) proximal convoluted tubule
   D) loop of Henle
   E) afferent arteriole

10. Which of the following is NOT a recognized oncogene/subcellular localization/gene product?
   A) ras/cell membrane/membrane associated G protein
   B) src/cell membrane/tyrosine kinase
   C) p53/nucleus/cell cycle-regulated nuclear transcription factor
   D) myc/nucleus/nuclear transcription factor
   E) myb/cell membrane/serine kinase

11. The following causal relationships between neoplasm and chemical cause are correct EXCEPT
A) hepatocellular carcinoma - aflatoxin B_1
B) basal cell carcinoma - diesel exhaust
C) pulmonary mesothelioma - asbestos
D) acute lymphoblastic leukemia - benzene
E) squamous cell carcinoma of the oral cavity - chewing tobacco

12. The supernatant fraction from a nuclear and cell debris pellet will contain
A) lysosomes
B) microsomes
C) NADPH-generating system
D) mitochondrial enzyme activity
E) all of the above

13. Which of the following serum enzyme tests would be most useful for detecting cytotoxic hepatocellular injury?
A) alkaline phosphatase and gamma-glutamyltranspeptidase
B) aspartate aminotransferase and sorbitol dehydrogenase
C) lactic dehydrogenase and glutamine oxaloacetic transaminase
D) all of the above
E) none of the above

14. People with glucose-6-phosphate dehydrogenase deficiency should avoid
A) fava beans
B) primaquine
C) methylene blue
D) all of the above
E) none of the above

15. Which of the following is the best test for cholestasis?
A) urinalysis
B) serum alanine aminotransferase activity
C) serum albumin level
D) serum alkaline phosphatase activity
E) serum cholinesterase activity

16. According to the 1993 ICH guidelines, which of the following procedures was considered to be the most sensitive indicator of the adverse effect on spermatogenesis?
A) mating of treated males with untreated females (i.e., effects on pregnancy rates)
B) good pathological and histopathological examination of testes
C) sperm analyses (sperm counts, sperm motility, sperm morphology)
D) weights of male reproductive organs
E) levels of male sex hormones

17. Increasing respiratory minute volume, as during exercise, would tend to have what effect on the response to sulfur dioxide exposure?
A) increase respiratory toxicity
B) decrease respiratory toxicity
C) no effect
D) impossible to predict without knowing the solubility in pulmonary surfactant
E) reduced systemic exposure

18. Which of the following causes a primary myelinopathy?
A) ethyl alcohol
B) methyl mercury
C) tri-*o*-cresylphosphate
D) hexachlorophene
E) tributyltin

19. Which of the following cells are associated only with acquired immunity?
A) polymorphonuclear leukocytes
B) fibroblasts
C) megakaryocytes
D) T and B lymphocytes
E) all of the above

20. Which of the following cell types is NOT involved in the mechanism of allergic contact dermatitis (Type IV delayed hypersensitivity)?
A) Langerhans' cells
B) $CD4^+$ T lymphocytes
C) mast cells
D) keratinocytes
E) none of the above

21. Which immunologic assay is NOT included in the National Toxicology Program Tier I immunotoxicity screening panel in rodents?
A) spleen and bone marrow cellularity
B) pluripotent stem cell activity
C) natural killer (NK) cell activity
D) antibody plaque forming response
E) histopathology

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

22. Which of the following statements about the nervous system is/are true?
1) sensory neuropathies are characterized by agitation
2) tests of conduction velocity have proved to be particularly suitable for detection of peripheral nerve damage because, in general, all axons of a nerve are damaged simultaneously
3) the principal basis for toxic responses involving immediate memory loss followed by gradual deterioration is likely to be cortical lesions
4) batrachotoxin depolarizes the nerve membrane and this effect is reversed by tetrodotoxin

23. Botulinum toxin
1) interferes with the release of acetylcholine from the axon terminal of motor nerves
2) requires internalization at the axon terminal to produce an effect
3) results in muscle paralysis
4) produces effect which may be reversed through neural sprouting at the terminal

24. Delayed neurotoxicity is caused by
1) tri-*o*-tolylphosphate
2) diisopropyl fluorophosphate
3) leptophos
4) diazinon

25. Which of the following mammalian cell mutation assays is/are capable of detecting chromosomal effects?
1) L5178Y/TK$^{+/-}$ mouse lymphoma assay
2) CHO/HGPRT assay
3) AS52/XPRT assay
4) V79/HGPRT assay

26. Which of the following is/are correct?
1) ethoxyethanol is spermatotoxic
2) female workers have an increased risk of spontaneous abortions due to methoxyethanol exposure
3) infants of female workers exposed during pregnancy to ethoxyethanol have an increased risk of congenital malformations
4) the adverse reproductive effects of ethoxyethanol have been attributed to the corresponding dihydroxy metabolite

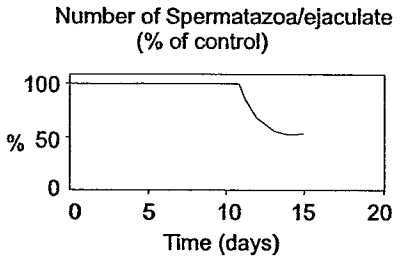
27. Which of the following structural variations or malformations is/are commonly associated with maternal toxicity?
1) hydronephrosis
2) fused or missing vertebrae
3) missing or supernumerary ribs
4) amelia or phocomelia

28. The loss of one or two base pairs in a gene is a
1) base pair substitution
2) frameshift mutation
3) translocation
4) point mutation

29. Which adverse effect(s) have been associated with high ambient airborne concentrations of particles in the range of 0.5 to 2.5 um?
1) asthmatic (bronchoconstrictive) episodes
2) premature death in the elderly
3) increased hospital admissions
4) childhood respiratory disease

30. Pulmonary fibrosis is caused by all of the substances in which of the following groups?
1) manganese, silica and cotton dust
2) asbestos, arsenic and nickel
3) coal dust, cotton dust and arsenic
4) asbestos, beryllium and silica

Figure 30007



31. A single dermal dose of a solvent was given to rats on day 3 in the figure above. Which of the following conclusions is/are consistent with the results shown?
1) fertility is decreased approximately 1 week after treatment
2) the effect on sperm production is consistent with the toxicity and slow dermal uptake and metabolism of 2-butoxyethanol
3) the results are consistent with the effect of 2-methoxyethanol on circulating levels of testosterone
4) the results are consistent with toxicity of the solvent to pachytene spermatocytes

32. The reactive airways dysfunction syndrome (RADS) is a chronic asthma-like condition associated with a single, high-dose exposure to a respiratory irritant. Which of the following has been reported to induce RADS?
1) toluene diisocyanate
2) chlorine gas
3) sulfur dioxide
4) ethylene oxide

33. The immunological mechanism of contact hypersensitivity includes which of the following?
1) migration of Langerhans cells to local lymph nodes
2) processing of foreign antigen by epidermal Langerhans cells
3) presentation of antigen by dendritic cells to local T lymphocytes
4) migration of T lymphocytes to the thymus

34. Which of the following is/are correct?
1) regulatory guidance on female reproductive function study design requirements for testing pesticides, therapeutic drugs, and food additives are the same
2) *fertility index* is calculated as the number of females conceiving as a percentage of the number of females mated with fertile males
3) decrements in overall group mean fertility are considered the most sensitive indicator of reproductive toxicology
4) a drug product carrying a reproductive risk of "X" from the US Food and Drug Administration (FDA) is contraindicated for use in women who are or may become pregnant

35. Endocrine disrupters may reasonably be expected to exert effects via the following mechanism(s)?
1) hormone mimicry
2) antagonism of binding to receptors
3) inhibition of enzymes
4) alteration of neuronal signaling pathways

36. Possible mechanisms that might result in thyroid tumor induction in rats is/are
1) inhibition of iodide transport in the thyroid
2) inhibition of 5'-monodeiodinase
3) inhibition of thyroperoxidase resulting in a disruption of T4 and T3 synthesis
4) increased TSH plasma concentrations resulting from an increased biliary excretion of T4 due to induction of hepatic UDP-glucuronyl transferase

37. Which of the following regarding allergic responses is/are correct?
1) there are four general categories of allergic responses: Types I, II, III and IV
2) Type I reactions are mediated by IgM antibodies and the main targets of this type of reaction are the GI tract, the skin, the respiratory system and the vasculature
3) Type IV reactions are also known as delayed hypersensitivity reactions and are mediated by sensitized T lymphocytes and macrophages
4) dose-response relationships are always apparent in allergic reactions

38. Pesticides that are actively secreted by the kidney include
1) paraquat
2) malathion
3) 2,4,5-trichlorophenoxyacetic acid
4) pentachlorophenol

39. Some microsomal enzyme inducers, including phenobarbital, nicardipine, spironolactone, chlordane and PCBs, disrupt thyroid gland function in rats. Which of the following is/are early distinguishing characteristics of the toxic action of these agents?
1) decreased thyroid hormone biosynthesis
2) plasma T3 and T4 are markedly decreased for a period of weeks
3) serum levels of T4-binding globulin are significantly increased
4) hepatic UPD-glucuronyltransferase activity is substantially increased

40. Trimellitic acid anhydride
1) is a widely used intermediate in paints, adhesives and coatings
2) like other acid anhydrides, has the potential to cause Type I hypersensitivity reaction characterized by asthma and rhinitis
3) with sufficient and repeated exposures, can produce Type III hypersensitivity reactions, characterized by influenza-like symptoms
4) may produce Type IV dermal sensitization

41. Inhibition of renal carbonic anhydrase leads to
1) metabolic alkalosis
2) excretion of alkaline urine
3) increased reabsorption of sodium ions in exchange for hydrogen ions throughout the nephron
4) a diuretic response with edema due to an increase in sodium bicarbonate and water excretion

42. Which of the following is/are correct?
1) metals, quinones, aldehydes, and acrylates are examples of haptens
2) the majority of skin sensitizers accept electron pairs from nucleophiles such as the amino and thiol groups found in cutaneous proteins
3) the induced allergenic response by a certain class of prohaptens requires metabolism or photoactivation
4) after dermal exposure, the epidermal Langerhans' cells encounter, recognize and bind allergens

43. Which of the following is/are correct?
1) the epidermis and pilosebaceous glands are the major sources of xenobiotic transformation in the skin
2) cytochrome P450 isozymes in the epidermis can be induced by certain chemicals upon exposure of the skin
3) human and rodent skin exhibit qualitatively similar Phase II reactions, but rodent skin often has a higher level of activity and isozyme expressions may differ across the two species
4) human epidermis contains sulfatases, beta-glucuronidase and certain reductases

44. Cataract formation has been noted in individuals with chronic exposure to
1) corticosteriods
2) diazepam
3) 2,4-dinitrophenol
4) glucuronic acid

45. Which of the following is/are considered major malformations, not variations or delays in development?
1) supernumerary ribs
2) exencephaly
3) incomplete ossification of sternebrae
4) spina bifida

46. Which of the following chemicals is/are considered nongenotoxic carcinogens?
1) d-limonene
2) chloroform
3) melamine
4) ethylene oxide

47. Which of the following anticoagulants inhibit(s) blood clotting by interference with the hepatic synthesis of vitamin K dependent clotting factors?
1) warfarin
2) heparin
3) dicoumarol
4) ancrod

48. Which of the following is/are true about the iris of the eye?
1) ocular penetration of very polar substances is poor
2) has double innervation to cause dilation and constriction
3) morphine acts centrally to constrict the pupil in the absence of a light stimulus
4) diisopropylfluorophosphate is an irreversible cholinesterase inhibitor that causes marked dilation

49. Which of the following is/are true?
   1) the stratum corneum prevents water loss from the underlying tissues
   2) occlusion of the skin with plastic wrap is a commonly employed technique to enhance the uptake of agents applied to the skin
   3) because rates of diffusion through the stratum corneum are slow, saturation of this structure can provide a depot, leading to continued penetration into the body for relatively long time periods after external exposure to an agent stops
   4) upon prolonged immersion in water, the stratum corneum can take up a great deal more water than its typical 20% hydration, and thereby increase its effectiveness as a barrier to agents with a hydrophobic character

50. Effects of aspirin in humans may include
   1) local GI hemorrhage
   2) alteration in uric acid excretion
   3) prevention of platelet aggregation
   4) inhibition of prostaglandin synthesis